

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

SYNTHETIC RUBBERS

MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	State synthetic form of NR and its structure.	MO1.01	R
2	Identify the monomer and preparation method of PBD.	MO1.01	U
3	State the structure and monomer names of IIR.	MO1.01	R
4	Identify the rubber used in oil seal.	MO2.04	A
5	What is the chemical name of CSM?	MO2.01	U
6	Specify name of any two specialty rubbers.	MO3.01	U
7	List out the two applications of silicone rubber.	MO2.04	A

8	What are the blends used in tyre tread?	MO4.04	U
9	The blend used in LPG tube.	MO4.04	A

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

Module Outcome Cognitive level

1	Differentiate any three properties of NR and IR	MO 1.04	R
2	Which are the monomers of EPDM rubber and narrate the function of diene monomer	MO 1.01	R
3	Differentiate the curing systems of NBR and CR	MO 2.03	U
4	Identify the structure and the reaction behind manufacturing of CSM	MO 2.01	U
5	How the ACN content effect on the properties of NBR?	MO 2.04	U
6	List out six characteristics of silicone rubber.	MO 2.04	U
7	Compare HNBR and XNBR.	MO 3.03	U
8	Distinguish TPE and blend.	MO4.01	U

9	List out any three advantages and disadvantages of TPE.	MO 4.01	U
10	List out the curing systems of IIR.	MO 1.03	U

PART C

III. Answer all questions from the following (6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	Illustrate the manufacturing of HOT emulsion SBR with flow chart.	MO 1.01	R
OR			
2	Explain the properties and application of IR Rubber	MO 1.04	U
3	Illustrate the manufacturing of EPDM with flow chart	MO 1.01	R
OR			
4	Illustrate the manufacturing of COLD emulsion SBR with flow chart	MO 1.01	U
5	Explain the properties and application of NBR	MO 2.04	U
OR			
6	Illustrate the manufacturing of CR rubber with flow chart	MO 2.01	U

7	Explain the structure, properties and application of CR RUBBER.	MO2.04	U
OR			
8	Explain the properties and applications of Halo butyl.	MO 2.01	U
9	Explain the manufacturing of silicone.	MO 3.01	U
OR			
10	Explain the properties and applications of HTPB.	MO 3.04	U
11	Explain the blending methods of rubber.	MO 4.03	U
OR			
12	Explain the properties and applications of TPU	MO 4.02	U

Prepared By:

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Lecturer

Department of Polymer Technology

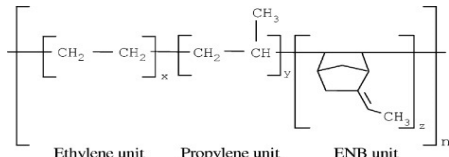
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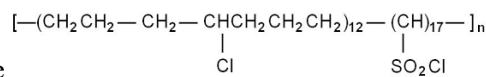
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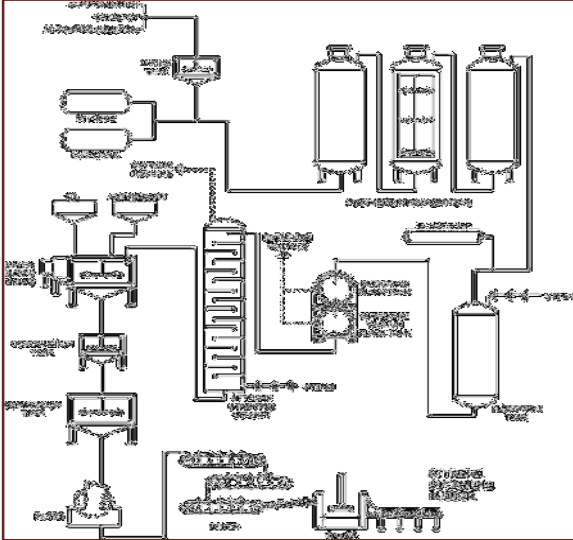
(Scoring Indicators)

Revision : 2021		Course Code :		
Course Title : SYNTHETIC RUBBER				
Qst. No	Scoring Indicator	Split up Score	Sub Total	Total
I	Answer all in one word or one sentence (1*9)			
1	Isoprene rubber	0.5		
	$\left[\text{CH}_2 - \text{CH} = \underset{\text{CH}_3}{\text{C}} - \text{CH}_2 \right]_n$	0.5	0.5*2	1
2	Butadiene monomer	0.5		1
	Solution polymerisation of butadiene	0.5	0.5*2	
3	Isobutylene and isoprene are the monomers of IIR	0.5		1
	Isobutylene Rubber (IIR)	0.5	0.5*2	1
	$\left(\text{CH}_2 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} \right)_m \left(\text{CH}_2 - \underset{\text{CH}_3}{\text{C}} = \text{CH} - \text{CH}_2 \right)_n$ Isobutylene Isoprene			
4	Rubber for oil seal-NBR	0.5	0.5*2	1
5		0.5		
	CSM-Chlorosulphonated poly Ethylene	0.5		
		0.5	0.5*2	1
6	Speciality rubbers-Silicone ,fluorocarbon,CSM	0.5*2		
		0.5*2	0.5*2	1

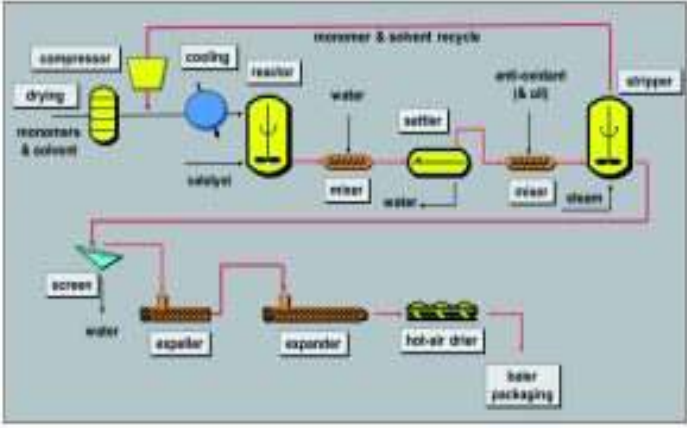
7	Application of silicone rubber-Contact lenses, electronic key pads,Gaskets,tubes ,Ice trays, spatula	0.5		
8	NR-SBR, NR-PBD,NR-SBR-PBD	0.5	0.5*2	1
9	LPG Tube-NBR/PVC	0.5		
		0.5	0.5*2	
II	PART B			
1	Each question carry 3 marks Answer any 8(3*8) Natural rubber consist almost entirely of the cis-1,4 poly isoprene, synthetic isoprene is a blend of cis-1,4, trans-1,4 and 3,4 vinyl polymer NR is either harvested in the form of latex from rubber trees, IR is synthetically produced by solution polymerization of 1-methyl-1,3-butadiene (Isoprene monomer) NR has more non rubber constituents So less purity IR has Less non rubber constituents So high purity	1 1 1	3	3
2	 One double bond of diene is used for polymerisation with EP ,while the other is remained in diene chains attached to polymer and this help the polymer to sulphur vulcanization ,it improves blend ability and reduce cost	1.5 1.5	3	3
3	NBR- sulphur +accelerator- EV system CR- Metal oxide zno,acid acceptor-Mgo,Ethylenethio urea-aaccelerator IIR-PF resin,leadoxide,sulphur	1 1 1	3	3
4	CSM is obtained by reacting polyethylene in solution with chlorine and sulphur dioxide	2	3	3



5	NBR with Lower ACN (increase ppts)	NBR with Higher ACN (increase ppts)	1		
	Processability Compression Set Sulfur	Cure Rate with Resilience	0.5*3	1.5*2	3
6	Oil/Fuel Resistance	Low Temperature Flexibility	0.5*3		
	Wide service temperature range – excellent thermal and thermoxidative resistance			3	3
7	Excellent resistance to attack by oxygen, ozone		1		
	Readily resistant to electromagnetic and particle radiation		1		
8	Excellent non-stick , Greater softness		1		
	non-toxic and food safe				
9	Low temperature flexibility due to their low glass transition temperature (Tg).				3
	HNBR- Hydrogenated Nitrile Rubber is the process of saturating the double bonds of the butadiene of NBR with hydrogen to give better resistance to oxidation and ozone and superior physical				
8	XNBR- Carboxylated NBR is the process of adding carboxylic acid groups to the nitrile backbone. These polymers will give improved tensile strength, tear and abrasion resistance.		1.5	1.5*2	3
	TPE- TPE has the properties and performance of rubber; it's actually processed like plastic and recyclable. Because of its elastomeric component, TPEs are soft and flexible materials				
9	BLEND- The most important concept in polymer blends is additivity of properties. when a polymer is mixed with another polymer, the resulting blend has average properties of the individual polymers		1.5	1.5*2	3
	Advantages of TPE - It can be scrapped and recycled,less energy consumption for production - Long lifespan - These have combined flexibility (rubber) with toughness (thermoplastics)		1.5		

10	Dis advantages of TPE TPEs have high cost compared to other plastics. If they are heated to a relatively high temperature, they tend to lose their 'rubbery' property. Tooling can be costly Sulphur +accelerator curing,PF resin curing,lead oxide curing	. 1.5 1.5 1+1+1	1.5*2 1.5*2	3
1	PART C Answer ALL questions each carry 7 marks(6*7)=42 1 Illustrate the manufacturing of hot emulsion SBR with fl  Butadiene-75 Styrene-25	2+1+4	2 1	7

2	Dodecyl Mercaptan(Activator)-0.5 Benzoyl Peroxide(Initiator)-0.5 Soap(Emulsifier)-5 Water-180 The monomers are continuously metered into the reactor chains and emulsified with the emulsifiers and catalyst agents. In hot polymerization, potassium peroxy disulphate is used as an initiator. Temperature of process-50°C pressure-0.5mpa. Mercaptan is added to control the molecular weight distribution It acts as a chain transfer agent to prevent the molecular weight. Polymerization is normally allowed to proceed to about 70% in hot polymerization before it is stopped with a shortstop agent diethyl hydroxylamine. Once the latex is properly short stopped, the unreacted monomers are stripped off the latex. Butadiene is stripped by degassing the latex by means of flash distillation and reduction of system pressure. Styrene is removed by steam stripping the latex in a column. The latex is then stabilized with the appropriate antioxidant and transferred to blend tanks. The latex is then transferred to finishing lines to be coagulated with sulphuric acid. The coagulated crumb is then washed, dewatered, dried, baled and packaged OR Properties- Less non rubber constituents So high purity, More uniformity in batches, Low Cis content so low mechanical properties tensile and hardness. But good tear resistance, tackiness and compression set ,Low initial viscosity, easy processability in mill, transparent ,No protein so non allergic Application- Tubing, dip-molded products, sports goods, medical equipment, protective gloves, conveyor lines, adhesives, sealants footwears, medical stoppers, sheets		4	
		4*1	4	
		3*1	3	7

3		2	2	
4	Polymerization of the ethylene, propylene and catalyst systems takes place in an excess of hydrocarbon solvent. The ethylene content of EPDM rubber is around 45% - 75%, Dienes used in the manufacture of rubbers are ethylenenorbornene (ENB), dicyclopentadiene (DCPD), and vinyl norbornene (VNB). - Stabilizers and oils are added directly after polymerization (if required). - Hot water, steam or mechanical devolatilization is used to flash the solvent and unreacted monomers, leaving only the EPDM compound. - The remaining crumb-like polymer is then dried by either mechanical press, drying ovens, or dewatering in screens OR	3 1 1 2+1+4	3 1*2 3	7 7

5	The diagram illustrates a chemical process flow. At the top, there are three large vertical storage tanks. Below them, a network of pipes and pumps connects various processing units. On the left, there are two smaller tanks and a pump. In the center, there is a tall vertical column. On the right, there is another large vertical tank. At the bottom, there is a horizontal unit that looks like a distillation or separation column, followed by more piping and a small pump. The entire system is enclosed in a rectangular frame. chart Butadiene-75 Styrene-25 REDOX (Initiator)-0.5 Soap(Emulsifier)-5 Water-180 The monomers are continuously metered into the reactor chains and emulsified with the emulsifiers and catalyst agents. In hot polymerization, potassium peroxy disulphate is used as an initiator. Temperature of process-5°c pressure-0.5mpa. Mercaptan is added to control the molecular weight distribution It acts as a chain transfer agent to prevent the molecular weight. Polymerization is normally allowed to proceed to about 70% in hot polymerization before it is stopped with a shortstop agent diethyl hydroxylamine. Once the latex is properly short stopped, the unreacted monomers are stripped off the latex. Butadiene is stripped by degassing the latex by means of flash distillation and reduction of system pressure. Styrene is removed by steam stripping the latex in a column. The latex is then stabilized with the appropriate antioxidant and transferred to blend tanks. The latex is then transferred to finishing lines to be coagulated with sulphuric acid. The coagulated crumb is then washed, dewatered, dried, baled and packaged	1*4	4	7
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6	Nitrile rubber is most popular for its oil resistance, although it's also tough and temperature resistant. The amount of oil resistance in nitrile rubber depends on the percentage of acrylonitrile. More acrylonitrile makes the rubber more resistant to swelling from oil, stronger, and less permeable to gases. Less acrylonitrile makes it more flexible at lower temperatures. It's also known for its heat resistance, strength, and resilience. The uses of nitrile rubber include disposable non-latex gloves, automotive transmission belts, hoses, O-rings, gaskets, oil seals, V belts, static & dynamic hydraulic seals, synthetic leather, printer's form rollers, and as cable jacketing. OR Oil-Based Neoprene Free radical emulsion polymerization is used exclusively today. It is carried out in a commercial scale using both batch and continuous processes With the aid of radical initiators, chloroprene in the form of an aqueous emulsion is converted into homopolymers or, in the presence of comonomers into copolymers. The polymerization is stopped at the desired conversion by the addition of a short	1+3 2+5	4 3 2 5	7 7
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7 8	stopping agent. The latex is freeze-coagulated on large, refrigerated revolving drums, from which it is drawn as a thin sheet. After washing and drying, the sheet is formed into a rope and then chopped to form the familiar chips (Picture 1) or granules. The physical, chemical and rheological properties of the different grades of commercial polychloroprene are dependent on the ability to change the molecular structure by changing polymerization conditions, e.g. polymerization temperature or monomer conversion, polymerization aids (comonomers, type and amount of molecular weight modifier and emulsifier) and conditions during finishing $\left[\text{CH}_2 - \underset{\text{Cl}}{\text{C}} = \text{CH} - \text{CH}_2 \right]_n$ PROPERTIES • Flame resistance electrical properties are good. • ,chemical resistance ,tackiness,Tensile strength APPLICATION • Flame resistant gaskets seals,diaphragms,bridge bearing pads ,cables,hoses,adhesives, OR PROPERTIES High physical strength, vibration damping, low glass transition temperature, low permeability, and resistance to aging and weathering from atmospheric exposure. It has broader vulcanization characteristics than butyl rubber due to the halogen reactivity. The halogenated butyl rubbers display better co-vulcanization properties with unsaturated general-purpose elastomers. SYNTHESIS-The manufacture of the bromobutyl rubber is a two-step process. First is the polymerization of isobutylene and isoprene to	1 3 1*3 2+3+2	2	7
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9	produce butyl rubber (IIR), which is then followed by bromination to form bromobutyl rubber. Butyl rubber (IIR) is prepared from high purity isobutylene and isoprene. The mechanism of polymerization consists of complex cationic reactions, the reaction solution must be neutralized with an alkaline solution such as sodium hydroxide. The solvent is then flashed off and stripped using steam and hot water. Calcium stearate is also added to prevent polymer agglomeration. The resultant polymer water slurry is screened, dried, and cooled in a process similar to that of regular (unhalogenated) butyl rubber. APPLICATION ▪ HIIR is mainly used in inner liners of tubeless tyres, adhesives, sealant etc	1		7
	• Silicone rubbers synthesis majorly involves three steps i.e. preparation of chlorosilanes followed by hydrolysis and then condensation polymerization yielding silicone elastomers 1. Preparation of chlorosilanes- The reaction giving chlorosilanes takes place in a fluidised bed of silicon metal powder in which flows a stream of methyltrichloride, usually at temperatures of 250 to 350°C and at pressures of 1 to 5 bars. A copper-based catalyst is used. The methyltrichlorosilane is separated via distillation and used as the monomer 2. The hydrolysis of methyltriichlorosilane in the presence of excess water form a trisilanol "$\text{Me}_2\text{Si}(\text{OH})_3$" Condensation -$\text{Me}_2\text{Si}(\text{OH})_3$ readily condenses, with HCl acting as a catalyst, to give a mixture of linear or cyclic oligomers . They must be condensed (linear) or polymerized (cyclic) and cross linking to obtain elastomers. OR	2*3		7
		1	1	7

10	$\text{HO}-\left[\begin{array}{c} \text{CH}_2-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_2 \\ \\ \text{CH} \\ \\ \text{CH}_2 \end{array} \right]_n-\text{OH}$	3	6	
11	PROPERTIES- Excellent hydrolytic stability, compatibility with oil-type plasticizers (coined as “oil extendibility”), improved adhesion to substrates, and improved low-temperature properties, with embrittlement at or below -80 °C. HTPB is a universally accepted binder, in view of its high C, H content and low oxygen mass (~5%), high solid loading capacity, low glass transition temperature (Tg -78-80 °C) and clean cross-linking reaction APPLICATION- An important application of HTPB is in solid rocket propellant. the oxidizing agent, fuel and other ingredients into a solid but elastic most composite propellant systems. The cured polyurethane acts as such mixtures.. HTPB also gained favor as the “Poly B-D Liquid R choice for many commercial applications, such caulks, foams, seal adhesives]. Five different methods are used for the preparation of polymer blends : melt mixing, solution blending, latex mixing, partial block or graft copolymerization, and preparation of interpenetrating polymer networks (IPN). An IPN is defined as a blend of two or more polymers in a network form, at least one of which is synthesized and /or cross-linked in the immediate presence of the other. Melt mixing is the most widespread method of polymer blend	1 1*6 1*3	7	

	preparation in practice. The blend components are mixed in the molten state in extruders or batch mixers. Solution blending is frequently used for preparation of polymer blends on a laboratory scale. The blend components are dissolved in a common solvent and intensively stirred. The blend is separated by precipitation or evaporation of the solvent. Latex blending- high quality dispersion of component materials .Blending is done in latex stage Partial block or graft copolymerization Dissolving or dispersing one polymer in the monomer of another and polymerising. OR It is a linear segmented block copolymer composed of hard and soft segments Soft Segment (polyether or polyester): It is built out of a polyol long-chain diol and an isocyanate which provides flexibility and elastomeric character of a TPU. Hard Segment (aromatic or aliphatic): It is constructed from a chain extender short-chain diol and isocyanate giving TPU its toughness and physical performance properties High abrasion and scratch resistance ensure durability and aesthetic value • Soft touch/high comfort • UV resistance, combination of high transparency with abrasion resistance • High elasticity across the entire hardness range • Excellent low-temperature and impact strength • Resistance to oils, greases and numerous solvents	1*4		7
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12	• Good flexibility over a wide temperature range • Robust weather and high-energy radiation resistance • automotive, wires and cables, breathable films for leisure, sports and textile coatings, weatherable, non-yellowing films, Seals and gaskets. Polymer Blending Generally elastomers are blended in one of 5 ways; 1. Latex blending 2. Solution blending 3. Combined solution / latex blending 4. Mechanical blending, and 5. Mechano-chemical blending Latex blending gives a high quality dispersion of component materials, with the blend potentially having lower hysteretic properties and lower solvent swelling. Examples of where such blending is found would be in natural rubber SBR latex blends. Solution blending of elastomers is normally done during the solution polymerization process. At the same time, carbon black and oils can be added to produce carbon black masterbatches. solution blending of incompatible polymers is difficult due to phase separation. Furthermore, rapid solvent evaporation is required to minimize heterogeneity which together restricts the applications of this technique. Combined solution and latex blending has an advantage in, for			
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	example, solutions of polymers mixed with aqueous suspensions. Carbon black migrates into the organic phase thereby giving a highly homogeneous dispe Polymer Blending Generally elastomers are blended in one of 5 ways; 1. Latex blending 2. Solution blending 3. Combined solution / latex blending 4. Mechanical blending, and 5. Mechano-chemical blending Latex blending gives a high quality dispersion of component materials, with the blend potentially having lower hysteretic properties and lower solvent swelling. Examples of where such blending is found would be in natural rubber SBR latex blends. Solution blending of elastomers is normally done during the solution polymerization process. At the same time, carbon black and oils can be added to produce carbon black masterbatches. solution blending of incompatible polymers is difficult due to phase separation. Furthermore, rapid solvent evaporation is required to minimize heterogeneity which together restricts the applications of this technique. Combined solution and latex blending has an advantage in, for example, solutions of polymers mixed with aqueous suspensions. Carbon black migrates into the organic phase thereby giving a highly homogeneous			
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Mark Distribution

Module	Hr/module	Marks/module	Type of Questions							
			Part A		Part B		Part C		Total	
			Number of questions	marks	Number of questions	marks	Number of questions	marks	Number of questions	marks
1	18	32	3	3	3	9	4	28	8	40
2	16	32	3	3	4	12	4	28	11	43
3	12	30	1	1	1	3	2	14	6	18
4	12	30	2	2	2	6	2	14	6	22
Total	58	124	9	9	10	30	12	84	31	123

Blue Print**Cognitive Level Mark Distribution**

Cognitive Level	Marks	% of Marks
Remembering	$2+6+14=22$	18
Understanding	$4+24+70=98$	80
Applying	3	2
Analysing		
Evaluating		
Total	123	100

Question Wise Analysis

Course: SR QP 1

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	MO 1.01	R	1	2
I.2	MO 1.01	U	1	2
I.3	MO 1.01	R	1	2
I.4	MO 2.04	A	1	2
I.5	MO 2.01	U	1	2
I.6	MO 3.01	U	1	2
I.7	MO 2.04	A	1	2
I.8	MO 4.04	U	1	2
I.9	MO 4.04	A	1	2
...II.1...	MO 1.04	R	3	7

.II 2...	MO 1.01	R	3	7
.II.3...	MO 2.03	U	3	7
. II.4...	MO 2.01	U	3	7
.II.5..	MO 2.04	U	3	7
. II.6...	MO 2.04	U	3	7
. II.7...	MO 3.03	U	3	7
.II.8..	MO4.01	U	3	7
II.9	MO 4.01	U	3	7
II.10	MO 1.03	U	3	7
III	MO 1.01	R	7	17
IV	MO 1.04	U	7	17
V	MO 1.01	R	7	17

VI	MO 1.01	U	7	17
VII	MO 2.04	U	7	17
VIII	MO 2.01	U	7	17
IX	MO 2.04	U	7	17
X	MO 3.01	U	7	17
XI	MO 2.01	U	7	17
XII	MO 3.04	U	7	17
XIII	MO 4.03	U	7	17
XIV	MO 4.02	U	7	17

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

SYNTHETIC RUBBERS

MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

IV. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	State the structure of isoprene rubber	MO 1.01	R
2	Identify the monomer of SBR.	MO 1.01	R
3	State the monomer names of EPDM	MO 1.01	R
4	Identify the rubber used in LPG TUBE and IB CAP	MO 2.04	A
5	HNBR is -----,XNBR IS -----	MO 3.03	U
6	Specify name of any two-diene used for EPDM	MO 1.01	U
7	List out the two applications of CSM	MO 2.04	U
8	What are the blends used Hawaichappel	MO 4.04	U
9	Which are the rubbers used in calculator key Pads, propellant binders	MO 3.04	A

PART B

V. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Differentiate the three properties of NR and IR	MO 1.04	R
2	State the structures of CSM, CR, NBR	MO 2.01	U
3	State the monomers of EPDM rubber, what is the function of diene monomer.	MO 1.01	R

4	List out the three properties of Halo butyl	MO 3.01	U
5	List out the four applications of CSM	MO 2.04	U
6	List out any three properties of Fluro rubbers	MO 4.01	U
7	Define TPE and blend	MO 4.02	U
8	Advantages of TPE	MO4.01	U
9	What is NVC	MO 4.04	U
10	What are the advantages of blends	MO 4.02	U

PART C

VI. Answer all questions from the following

(6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	Explain the Structureapplication and properties of IR	MO 1.04	U
OR			
2	Manufacturing of cold SBR with flow chart	MO 1.01	U
3	Explain the structure properties andapplication of PBD	MO 1.01	U
OR			
4	Explain the manufacturing of EPDM with flow chart	MO 1.01	U
5	Explain the manufacturing ofNBR with flow chart	MO 2.01	U
OR			
6	Explain the structure, properties, application ofsilicone	MO 2.04	U

7	Explain the properties, application of CSM	MO 2.04	U
OR			
8	Explain the manufacturing of CR with flow chart	MO 2.01	U
9	Explain the properties, application of acrylic rubbers	MO 3.01	U
OR			
10	Explain the structure, properties, application of HTPB	MO 3.03	U
11	Explain the properties and application of NR-SBR,NR -BR	MO 4.03	U
OR			
12	List out Advantages and disadvantages of TPE	MO 4.02	U

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SCHEME OF VALUATION

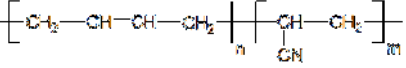
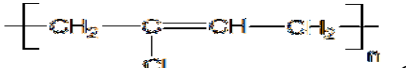
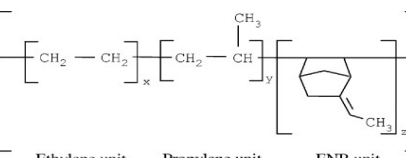
(Scoring Indicators)

Revision : 2021 Course Code :

Course Title : SYNTHETIC RUBBERS

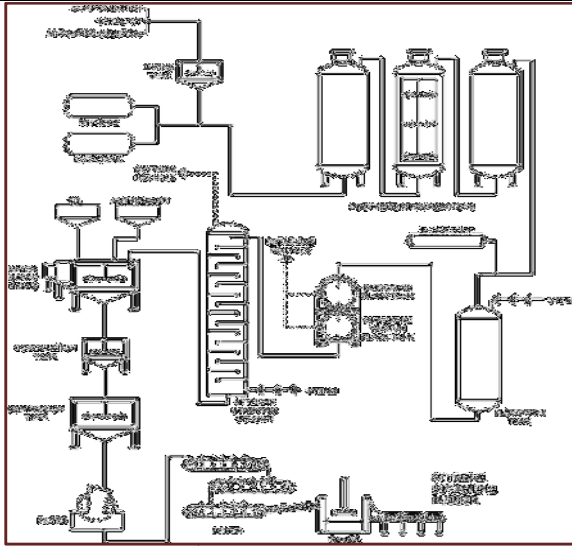
Qst. No	Scoring Indicator	Split up Score	Sub Total	Tota l
I	Answer all in one word or one sentence (1*9)			

1	Isoprene rubber $\left[\text{CH}_2 - \underset{\text{CF}_3}{\text{CH} = \text{C}} - \text{CH}_2 \right]_n$	1	1	1
2	Styrene and butadiene	0.5*2	1	1
3	ETHYLENE PROPLENE DIENE MONOMER	1	1	1
4	LPG- NBR,IB Cap-IIR	0.5*2	1	1
5	Hydroginatednitrilerubber,carboxilated nitrile	0.5*2	1	1
6	Ethylene norbonene,dicyclopentadiene	0.5*2	1	1
7	Cable, coatings NR-HSR,NR-EVA	0.5*2	1	1
8	SILICONE,HTPB	0.5*2	1	1
9				
II	PART B			
	Each question carry 3 marks Answer any 8(3*8)			
1	Natural rubber consist almost entirely of the cis-1,4 poly isoprene, synthetic isoprene is a blend of cis-1,4, trans-1,4 and 3,4 vinyl polymer NR is either harvested in the form of latex from rubber trees, IR is synthetically produced by solution polymerization of 1-methyl-1,3-butadiene (Isoprene monomer) NR has more non rubber constituents So less purity IR has Less non rubber constituents So high purity	1+1+1	3	3
2	$\left[\text{---}(\text{CH}_2\text{CH}_2\text{---CH}_2\text{---}\underset{\text{Cl}}{\text{CH}}\text{CH}_2\text{CH}_2\text{CH}_2)_n\text{---}(\text{CH})_{17}\text{---} \right]_n$ SO₂Cl CSM	1+1+1		

	 NBR  CR  Ethylene unit Propylene unit ENB unit		3	3
3	One double bond of diene is used for polymerisation with EP ,while the other is remained in diene chains attached to polymer and this help the polymer to sulphur vulcanization ,it improves blend ability and reduce cost Co vulcanization property, flame resistance ,improve blendability,curing versatility,	1+2	3	3
4	Roof and chemical tank Lining, coating,chemical resistant valves mouldings,cable,hose covers	1+1+1	3	3
5	High temperature resistance, chemical resiatance,weather resistant,	1+1+1		
6		1+1+1	3	3
7	TPE- TPE has the properties and performance of rubber; it's actually processed like plastic and recyclable. Because of its elastomeric component, TPEs are soft and flexible materials BLEND- The most important concept in polymer blends is additivity of properties. when a polymer is mixed with another polymer, the resulting blend has average properties of the individual polymers	1+1+1	3	

8	Advantages of TPE • It can be scrapped and recycled,less energy consumption for production • Long lifespan • These have combined flexibility (rubber) with toughness (thermoplastics)	1+1+1 1+1+1	3	3
9	NVC is a TPE,It is produced by physical blending of NBR+PVC ,highly oil resistant and flame resistant rubber ,good tackiness Achieve superior properties than invidauual,reduce cost, improve processabilty and vulcanisation versatility	1+1+1	3	3
10	PART C Answer ALL questions each carry 7 marks(6*7)=42 Properties- Less non rubber constituents So high purity, More uniformity in batches,LowCis content so low mechanical properties tensile and hardness. But good tear resistance, tackiness and compression set ,No protein so non allergic Application-hospital sheetings, gloves,gaskets,washers	1+1+1	3	3
1	$\left[\text{CH}_2 - \text{CH} = \underset{\text{CH}_3}{\text{C}} - \text{CH}_2 \right]_n$ OR Illustrate the manufacturing of COLD emulsion SBR with flow chart	1+1+1 3 3 1	3+3+1	7

2



2+5

7

The monomers are continuously metered into the reactor chains and emulsified with the emulsifiers and catalyst agents. In hot polymerization, potassium peroxy disulphate is used as an initiator. Temperature of process-5°C pressure-0.5mpa.

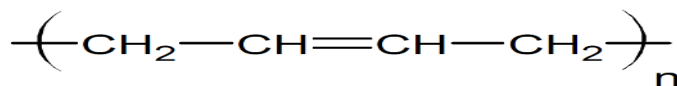
Mercaptan is added to control the molecular weight distribution. It acts as a chain transfer agent to prevent the molecular weight. Polymerization is normally allowed to proceed to about 70% in hot polymerization before it is stopped with a shortstop agent diethyl hydroxylamine.

Once the latex is properly short stopped, the unreacted monomers are stripped off the latex. Butadiene is stripped by degassing the latex by means of flash distillation and reduction of system pressure. Styrene is removed by steam stripping the latex in a column.

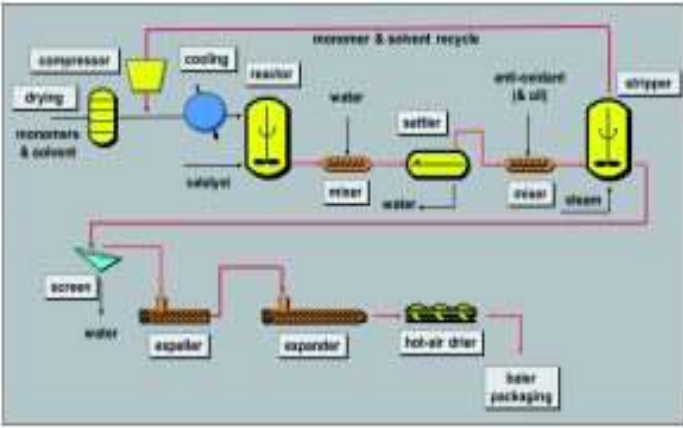
The latex is then stabilized with the appropriate antioxidant and transferred to blend tanks. The latex is then transferred to finishing lines to be coagulated with sulphuric acid. The coagulated crumb is then washed, dewatered, dried, baled and packaged

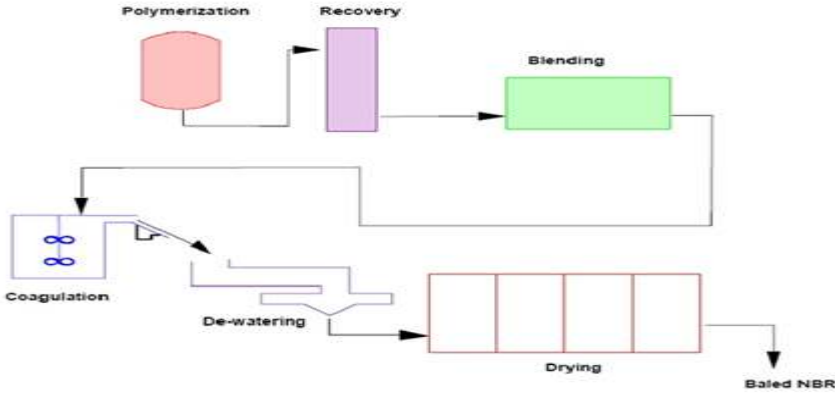
3

Polybutadiene (BR)



1+3+3

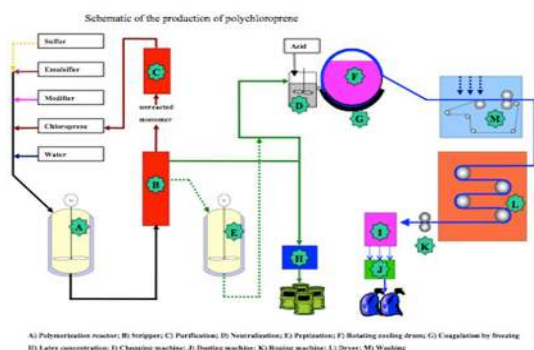
4	Properties-High resilience,Abrasion resistance, cut growth resistance, cold flow, low weather resistance Application-tyre tread, conveyer belt, golf ball, hose OR 	2+5		7
5	Polymerization of the ethylene, propylene and catalyst systems takes place in an excess of hydrocarbon solvent. The ethylene content of EPDM rubber is around 45% - 75%,Dienes used in thE manufacture of EPDM rubbers are ethylidene norbornene (ENB), dicyclopentadiene (DCPD), and vinyl norbornene (VNB). - Stabilizers and oils are added directly after polymerization (if required). - Hot water, steam or mechanical devolatilization is used to flash the solvent and unreacted monomers, leaving only the EPDM compound. - The remaining crumb-like polymer is then dried by either mechanical press, drying ovens, or dewatering in screens			7

		2+5		
6	▶ NBR is produced in an emulsion polymerization system. ▶ The water, emulsifier/soap, monomers (butadiene and acrylonitrile), radical generating activator, and other ingredients are introduced into the polymerization vessels. ▶ The emulsion process yields a polymer latex that is coagulated using various materials (e.g. calcium chloride, aluminum sulfate) to form crumb rubber that is dried and compressed into bales . ▶ Different polymerization temperatures to make "hot" and "cold" polymers. ▶ Cold polymers are polymerized at a temperature range of 5 to 15°C. The lower polymerization temperatures yield more-linear polymer chains. ▶ Hot NBR polymers are polymerized at the temperature range of 30 to 40°C. This process yields highly branched polymers with good tack and hot tear strength The hot polymers' have natural resistance to flow Hot NBR is mainly used in adhesives and molded products OR • Wide service temperature range – excellent thermal and thermoxidative resistance • Excellent resistance to attack by oxygen, ozone and sunlight • Readily resistant to electromagnetic and particle radiation (UV, alpha, beta and gamma rays) • Excellent non-stick , Greater softness		7	
7				

8	• E&E applications of silicone rubbers are cable terminations, connectors, tubes, cables, insulators, parts for energy transmission, engine gaskets, control unit gaskets etc. They are also widely used in lighting applications such • as lens molds, reflectors. Calculator and phone key pads & covers, camera gaskets, silicone heaters • Medical applications are catheters, tubing, drainage seals, gaskets, o-rings, valves, membranes, respiratory care-e. g. masks, medical instruments, medical cables, medical textile coating, baby bottle nipples,artifical implants and more. • Food industry valves ,gaskets,tubes,hoses of food processing as well as packaging equipments, drinks wending machines, baking trays and molds, oven safe pots, ice trays,spatula,brushes	3.5+3.5	7
	PROPERTIES • Ozone and weathering resistance is also excellent and electrical properties are good. • Good flame retardancy,chemical resistance • Low temperature flexibility and oil resistance is similar to chloroprene. • High Mooney viscosity • Good solubility and Low solution viscosity • Typical working temperature range: -20°C +150°C APPLICATION • Cable coverings, Anti-corrosion lining of chemical facilities, chemical equipments and pipelines. • High performance automotive hoses, special hose, pump hoses, hydraulic hose • Radiation shield, high rigidity products, water resistant roof seal • Hard coatings, adhesives, floor adhesives, outdoor architectural coatings	3.5+3.5	

OR

Oil-Based Neoprene



2+5

9

Free radical emulsion polymerization is used exclusively today. It is carried out in a commercial scale using both batch and continuous processes. With the aid of radical initiators, chloroprene in the form of an aqueous emulsion is converted into homopolymer or, in the presence of co monomers into copolymers. The polymerization is stopped at the desired conversion by the addition of a short stopping agent. The latex is freeze-coagulated on large, refrigerated revolving drums, from which it is drawn as a thin sheet. After washing and drying, the sheet is formed into a rope and then chopped to form the familiar chips (Picture1) or granules. The physical, chemical and rheological properties of the different grades of commercial polychloroprene are dependent on the ability to change the molecular structure by changing polymerization conditions, e.g. polymerization temperature or monomer conversion, polymerization aids (co monomers, type and amount of molecular weight modifier and emulsifier) and conditions during finishing.

Properties of acrylic rubber

Very good weather, ozone and hot air resistance.

Moderate strength.

Low elasticity.

Thermal stability from -30 to +150°C, up to +160°C for short periods.

Relatively poor low-temperature performance.

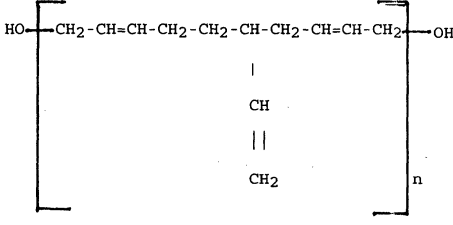
Good resistance to mineral oils.

Moderate resistance to fuels.

10

7

7

	Average rebound resilience. It is commonly used in automotive transmissions and hoses. It is also used in shaft seals, adhesives, beltings, gaskets and O-rings. It is used in vibration damping mounts due to the damping properties. OR  PROPERTIES- Excellent hydrolytic stability, compatibility with oil-type plasticizers (coined as “oil extendibility”), improved adhesion to substrates, and improved low-temperature properties, with embrittlement at or below -80 °C. HTPB is a universally accepted binder, in view of its high C, H content and low oxygen mass (~5%), high solid loading capacity, low glass transition temperature (Tg - 78-80 °C) and clean cross-linking reaction APPLICATION- An important application of HTPB is in solid rocket propellant. It binds the oxidizing agent, fuel and other ingredients into a solid but elastic mass in most composite propellant systems. The cured polyurethane acts as a fuel in such mixtures.. HTPB also gained favour as the “Poly B-D Liquid Resin” of choice for many commercial applications, such caulks, foams, sealants, and adhesives]. NR-BR BLENDOne particular advantage of blending BR with NR is the effect of high curing temperatures on physical properties. It s heat stability is conveyed by Cis BR when blended with NR .The addition of Cis BR is said to decrease the tensile strength and modulus of NR	3.5+3.5		
11	Excellent hydrolytic stability, compatibility with oil-type plasticizers (coined as “oil extendibility”), improved adhesion to substrates, and improved low-temperature properties, with embrittlement at or below -80 °C. HTPB is a universally accepted binder, in view of its high C, H content and low oxygen mass (~5%), high solid loading capacity, low glass transition temperature (Tg - 78-80 °C) and clean cross-linking reaction	1+3+3		
12	An important application of HTPB is in solid rocket propellant. It binds the oxidizing agent, fuel and other ingredients into a solid but elastic mass in most composite propellant systems. The cured polyurethane acts as a fuel in such mixtures.. HTPB also gained favour as the “Poly B-D Liquid Resin” of choice for many commercial applications, such caulks, foams, sealants, and adhesives]. NR-BR BLENDOne particular advantage of blending BR with NR is the effect of high curing temperatures on physical properties. It s heat stability is conveyed by Cis BR when blended with NR .The addition of Cis BR is said to decrease the tensile strength and modulus of NR	3.5+3.5		7

	Vulcanisate but to increase elasticity and abrasion resistance. The poor processing properties of Cis BR are improved by blending with NR. Application-truck Tyre tread, conveyer belt NR-SBR BLEND- Tread wear and wet skid resistance of SBR are found to improve for NR/SBR elastomer blends. The SBR component enhances toughness and NR provides superior resilience and energy return when they are blended. Application –passenger car Tyre tread, footwear, hose OR • Able to be scrapped and recycled, less energy consumption for production • Long lifespan • Provides both slip resistance and shock absorption • These have combined flexibility (rubber) with toughness (thermoplastics) • They have good tensile strength and are tear resistance. • They resist chemicals and ink / paint adhere well to them. • They have good resistance to compression. After bending, they will return close to their original shape / form • They can be extruded, blow moulded and injection moulded, easier molding ; shorter molding cycles; less expensive molding process; • MAIN DISADVANTAGES	3.5+3.5		7
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- | | | | | |
|--|---|--|--|--|
| | <ul style="list-style-type: none">• The main disadvantage of TPEs is their high cost compared to other plastics.• If they are heated to a relatively high temperature, they tend to lose their ‘rubbery’ property.• TPEs tend to ‘creep’. This is a tendency to move and deform when under the influence of stress, such as sustained pressure and temperature.• Tooling can be costly | | | |
|--|---|--|--|--|

Blue Print

Mark Distribution

Module	Hr/module	Marks/module	Type of Questions							
			Part A		Part B		Part C		Total	
			Number of questions	marks	Number of questions	marks	Number of questions	marks	Number of questions	marks
11 1	18	32	4	4	2	6	4	28	10	38
2	16	32	2	2	2	6	4	28	9	38
3	12	30	1	1	1	3	2	14	5	19
4	12	30	1	1	5	15	2	14	7	30
Total	58	124	9	9	10	30	12	84	31	123

Blue Print

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	9	7
Understanding	114	91
Applying	2	2
Analysing		
Evaluating		
Creating		
Total	123	100

Question Wise Analysis

Course : SYNTHETIC RUBBER QP 2

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	MO 1.01	R	1	2
I.2	MO 1.01	R	1	2
I.3	MO 1.01	R	1	2
I.4	MO 2.04	A	1	2
I.5	MO 3.03	U	1	2
I.6	MO 1.01	U	1	2
I.7	MO 2.04	U	1	2
I.8	MO 4.04	U	1	2
I.9	MO 3.04	A	1	2
...II.1...	MO 1.04	R	3	7
.II 2...	MO 2.01	U	3	7
.II.3...	MO 1.01	R	3	7

. II.4...	MO 3.01	U	3	7
.II.5..	MO 2.04	U	3	7
. II.6...	MO 4.01	U	3	7
. II.7...	MO 4.02	U	3	7
.II.8..	MO4.01	U	3	7
II.9	MO 4.04	U	3	7
II.10	MO 4.02	U	3	7
III	MO 1.04	U	7	17
IV	MO 1.01	U	7	17
V	MO 1.01	U	7	17
VI	MO 1.01	U	7	17
VII	MO 2.01	U	7	17
VIII	MO 2.04	U	7	17
IX	MO 2.04	U	7	17

X	MO 2.01	U	7	17
XI	MO 3.01	U	7	17
XII	MO 3.03	U	7	17
XIII	MO 4.03	U	7	17
XIV	MO 4.02	U	7	17